

Diana is currently 32 years old and plans to work until age 62, giving her approximately 30 remaining working years. She is considering two education options to increase her future earnings: a one-year Cyber Security Certification with a total cost of \$12,500, and a five-year evening Master's in Management program with annual tuition of \$55,000. To evaluate these options, a financial analysis was completed using a discount rate of 3.75%, which reflects the return available from a 5-year GIC. The purpose of this analysis is to determine which education option provides the greater financial benefit over Diana's remaining career.

Based on the financial analysis, both the Cyber Security Certification and the Master's in Management are financially worthwhile investments, as each program results in a positive net present value over Diana's remaining working years. However, the Cyber Security Certification is the stronger financial choice if Diana must select only one program.

The Cyber Security Certification requires a much smaller upfront investment of \$12,500 and is completed in one year. As a result, the associated salary increase begins in Year 2, allowing Diana to benefit from higher earnings much earlier in her career. In contrast, the Master's in Management requires tuition payments of \$55,000 per year over five years, significantly increasing the upfront cost and delaying the salary benefits until after program completion.

If the programs are not mutually exclusive, Diana could reasonably pursue both options. Completing the certification first allows her to increase her earnings sooner, potentially making it easier to afford the Master's program later. However, if only one program can be chosen, the Cyber Security Certification provides a faster and more efficient return on investment.

The financial calculations strongly support this recommendation. The present value of the salary differential from the Cyber Security Certification is approximately **\$394,848**, compared to a program cost of only **\$12,500**, resulting in a very high net benefit. The salary increase begins one year after program completion and continues to grow at an annual rate of 2.4% until retirement.

The Master's in Management generates a higher present value of salary increases, estimated at **\$510,525**, reflecting the larger base salary increase of \$30,000 growing at 2.3% annually. However, this benefit comes at a significantly higher cost. The present value of tuition payments alone is approximately **\$255,826**, which greatly reduces the overall net gain and increases financial risk.

Although the Master's program produces a higher total salary benefit, the certification offers a substantially better return relative to its cost, lower financial risk, and quicker recovery of the

initial investment. These results clearly indicate that the Cyber Security Certification is the preferred financial option under a single-choice scenario.

Hi Supervisor,

I have completed the financial analysis for Diana Wells' education options using a 3.75% discount rate based on current 5-year GIC returns. Both the Cyber Security Certification and the Master's in Management program produce positive NPVs and are financially beneficial when considered individually.

If the programs are not mutually exclusive, completing both would be financially advantageous over Diana's remaining working years. However, if only one option can be pursued, the Cyber Security Certification is the stronger financial choice due to its lower cost, higher relative return, and quicker payoff period.

Hi Diana,

Thank you for allowing us to review your education options. Using a 3.75% discount rate that reflects the return you could earn from a 5-year GIC, both the Cyber Security Certification and the Master's in Management program show positive financial value over your career.

If it is possible to complete both programs, doing so would increase your lifetime earnings. However, if you must choose only one, the Cyber Security Certification appears to be the better financial option. It requires a much smaller upfront investment, increases your salary sooner, and provides a quicker return on your money.